



Using the Multidrop

Contact the Automation Group for help or more info
automation@calicolabs.com

Multidrop Overview



- The Multidrop Combi is an **eight-channel microplate reagent dispenser** that uses a peristaltic pump to dispense liquids quickly and precisely into microplates
- Dispense volume range is **0.5uL to 2500uL**
- It accommodates **96, 384 and 1536-well plates** with plate heights from 5 to 50 mm
- 6 – 48 well plates as well as tubes in a 96-channel rack can be used
- Can be used, for example, for drug discovery/high throughput screening, genomic, proteomic and cell-based assays, and ELISA

Standard cassette:

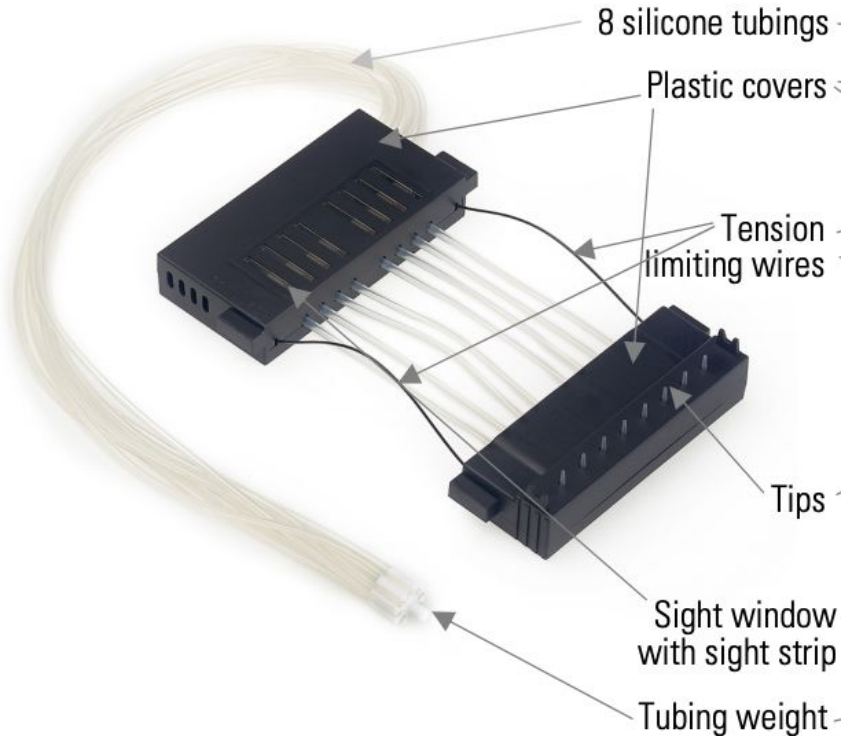
5uL-2.5 mL

Small cassette:

0.5uL to 50 µL



Multidrop Overview



- Uses detachable and autoclavable dispensing cassettes. The dispensing cassette contains eight silicone tubes.
- Each reagent can have a separate dispensing cassette to avoid mixing of reagents
- The same dispensing cassette can be used repeatedly with several reagents if it is washed in between uses

Additional Resources

To see the Multidrop in action, watch the videos here:

- https://www.youtube.com/watch?v=SEgmJkDt_tw
- <https://www.youtube.com/watch?v=TnqR10XoujE>

The full user manual is available [here](#).



Using the Multidrop

Follow the steps below to dispense reagents using the Multidrop:

- [Install the cassette](#)
- [Set up the dispense](#)
- [Save a Protocol](#)
- [Run a Saved Protocol](#)
- [Dispense](#)
- [Cleanup and Shutdown](#)



Install the Cassette

1.



Take the lower part of the cassette with the tip protection into your right hand with the dispensing cassette tips pointing down and the upper part into your left hand

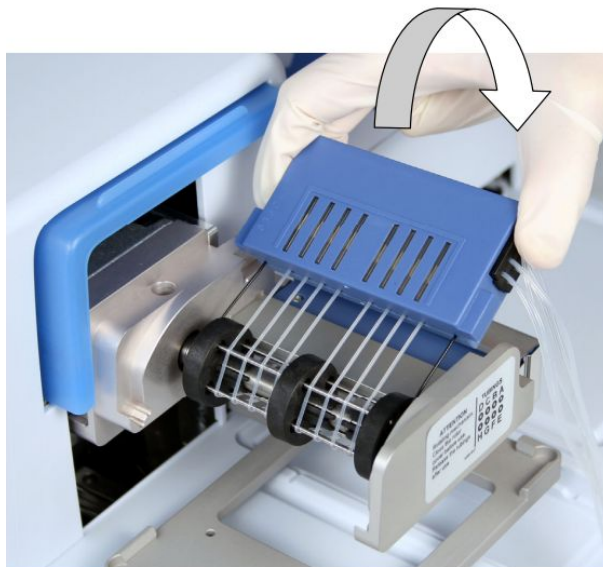
2.



Carefully place the eight tubes below the pump rotor and insert the lower part of the dispensing cassette into the lower part slots of the pump body

Install the Cassette

3.



Grip the upper part of the dispensing cassette and carefully pull the tubes around the pump rotor until the upper part reaches and fits into the upper part slots of the pump body. Insert into the slots.

4.



Double-check that both the upper and the lower parts of the dispensing cassette are properly placed into their corresponding slots. Pull the tip protection off from the dispensing cassette.

Install the Cassette

5.



Pull the rotor cover over the rotor. The dispense will not run unless the cover is closed.

6.

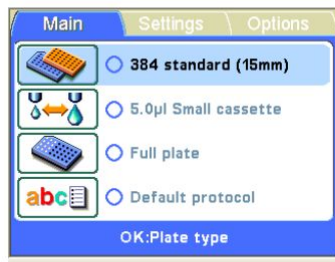


Pull the tubing out gently from the tubing protector cover and place the tubing into the reagent container. Make sure the tubing is sitting at the bottom of the container and that there is enough liquid in the container to submerge the base of the tubing.



Set up the Dispense using the Software

1 – **Plate type**; see Step 3 on p. 36.



If the plate type is correct, press

If you want to change the plate type, press

Select the plate type with

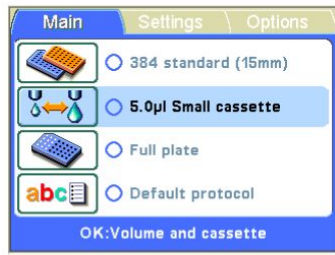
To accept, press



When the instrument is on, the screen will show the Main tab with the options shown in the images.

- Choose the correct plate type
- Choose the “standard” cassette type
- Set the volume to dispense (volumes can vary across columns, refer to [this](#) slide to see how)

2 – **Volume and cassette**; see Step 4 on p. 36.



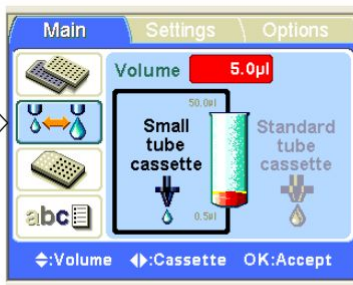
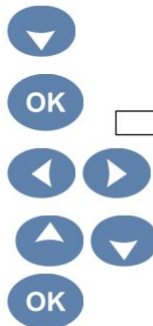
If the volume and the cassette are correct, press

If you want to change the volume, press

Select the appropriate cassette with

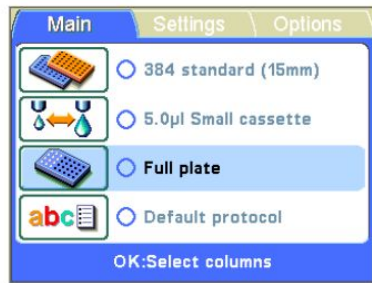
Set the needed volume with

To accept, press



Set up the Dispense using the Software

3 – **Columns**; see Step 5 on p. 36.



If you intend to fill the whole plate, you can start dispensing after priming the instrument.

If you intend to fill only part of the columns, press



To select the columns, navigate with



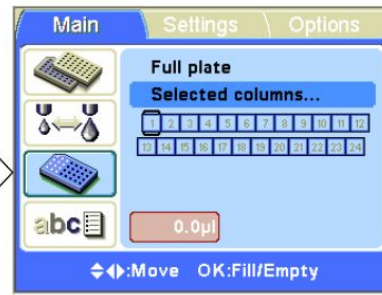
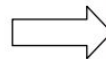
and to confirm each selection, press



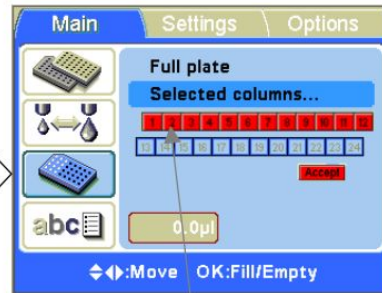
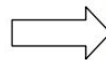
Last, select **Accept** with



When the **Accept** box is highlighted, press



Choose which columns to dispense into by clicking on the “full plate” option on the main screen and then selecting the desired columns



Selected column(s) = **red**



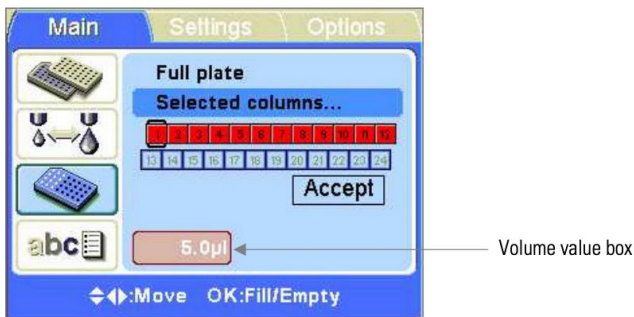
Set up the Dispense using the Software

Plate type	Cassette type	Max. dispensing volume * (µl)	Dispensing height (mm) **
96-well standard (15 mm)	Standard	500	16
96-well low DW (deep well) (22 mm)	Standard	600	23
96-well DW (44 mm)	Standard	2500	45
384-well low volume (7.5 mm)	Small	25	8.5
384-well low profile (10 mm)	Small	70	11
384-well standard (15 mm)	Small	130	16
384-well DW (22 mm)	Standard	200	23
384-well DW (44 mm)	Standard	300	45
1536-well low volume (5 mm)	Small	2	6
1536-well standard (10.5 mm)	Small	13	11.5

- The table here lists the volume that can be dispensed into each type of plate
- We only have standard cassettes at Calico (but small cassettes can be ordered)
- The default dispense height is 1mm above the height of the plate to avoid physical contact with the plate



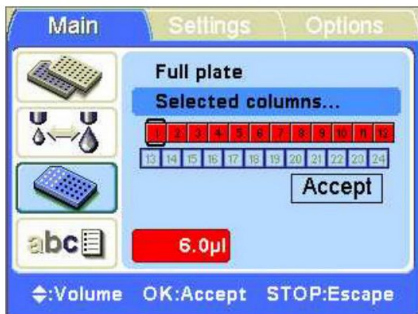
Set up the Dispense using the Software



Dispense volumes in each column can be customized to be different using the instructions on the left



Press the **OK** button for 0.5 seconds. The **Volume value** box is now activated (bright red). Adjust the dispensing volume of the column where the cursor is located using the **Up** and **Down** arrow keys.



Press **OK** after selecting the desired volume.



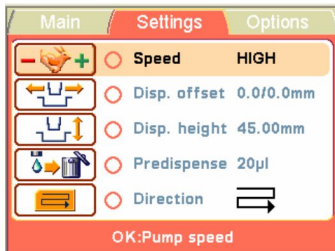
Set up the Dispense using the Software

Dispensing speed

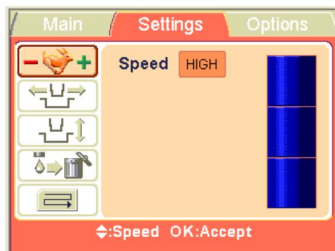


The default dispensing speed is *High*, but you can change the dispensing speed, for example, to reduce foaming. Note that the *Medium* dispensing speed gives the best dispensing results with most of the buffers.

Go to the **Settings** menu. Select the **Pump speed** row.



OK Press OK.



Use the **Up** and **Down** arrow keys to select the pump speed. The available settings are *High*, *Med* (medium) and *Low*. The default speed is *High*.

OK Press OK.

Dispensing height



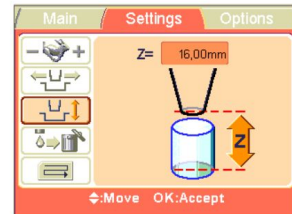
The default dispensing height depends on the plate type (see Table 4-3). You can change the dispensing height values if the plate used is not the standard one specified in the plate type list.

Go to the **Settings** menu. Select the **Dispensing height** row.



OK Press OK.

The track mechanism and pump body move to the setting position.



Use the **Up** and **Down** arrow keys to set the dispensing height. While you change the height, the pump body moves accordingly. The height is set in 0.05 mm increments in both the up and down directions. The minimum height value depends on the selected plate type (see Table 4-3). The default dispensing height is 1 mm above the selected plate.

OK

Press OK. If the OK button is not pressed within 60 seconds, the adjustments are disregarded and the main view returns to the **Settings** menu.

Saving a Protocol

Save and edit protocol

Go to the Main menu. Select the **Save and Edit Protocol** row.



OK Press OK.



▶ Press the **Right** arrow key to name/edit the name of your protocol.



Enter the name of your protocol.

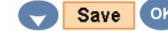


Use the arrow keys to choose the character you want.

OK Press OK to enter the said character.

Frequently used protocols can be saved and reused in the future.

- Enter the setting for the protocol using the steps from the previous slides
- Save the protocol immediately by going into the “default protocol” option on the screen
- Name and save the protocol



Save OK

Use the **Down** arrow key to select the **Save** button and then press the OK button to save the edited protocol name.

The sandglass window will appear while the Save action is in progress.



The protocol name is now shown on the **Save and Edit Protocol** row.



Running a Saved Protocol

Select protocol



To select an existing protocol for a particular plate type, first select the plate type (see “Plate type” on page 38) and then select the **Save and Edit Protocol** row. The protocols are listed according to each plate type.



OK

Press OK.



Select the protocol using the **Up** and **Down** arrow keys.

OK

Press **OK** to accept the selection.

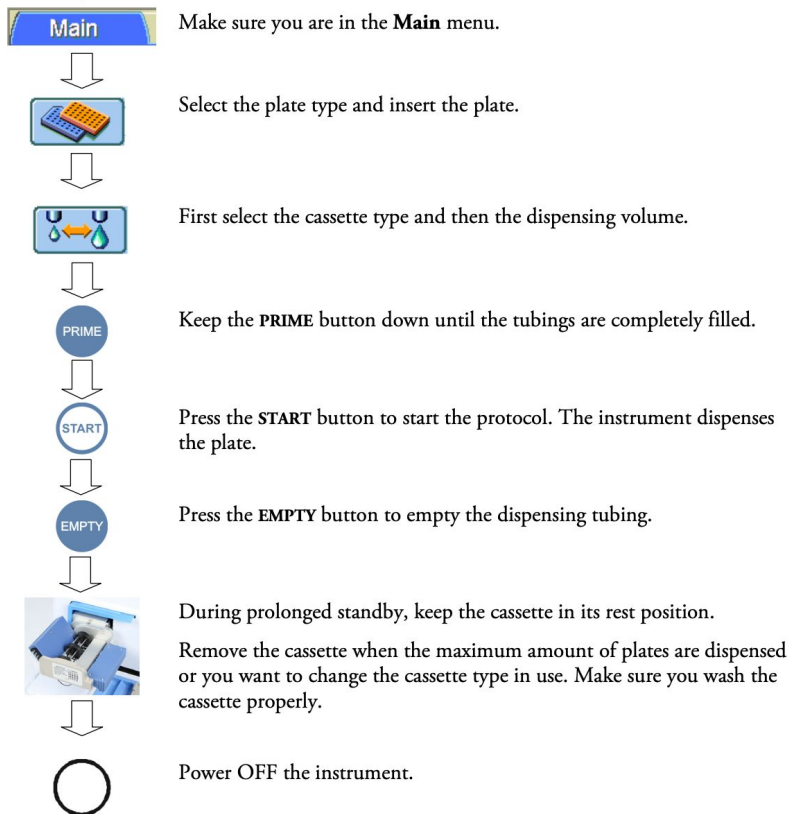
The protocol is now selected.



Note In the protocol list only the protocols made for the plate type in question are shown. ▲



Run the Dispense



- Install the cassette
- Place tubing into reagent reservoir
- Hold the Prime button down to allow reagent to fill the tubing
- Place receiving labware into the labware slot on the instrument
- Find or create and run the desired protocol
- Hold down the Empty button to retrieve reagent from the tubing back into the reservoir



Cleanup and Shutdown

- After emptying the tubing of reagent, place tubing into a reservoir of clean water and prime water through the tubing for 30 seconds to ensure that all reagent has been washed out of the tubing
- Remove the cassette and place tip protector back onto the cassette tips
- Wipe down any reagent residue droplets on the instrument

Questions/Concerns?

Please contact the automation team at automation@calicolabs.com

